Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
B		2607259

Beyond the Sky: A Dual-Modal Logistics Strategy for Lunar Colonization

Abstract:

To support the expansion of human civilization to a 100,000-person lunar colony, we established a multi-dimensional logistics and sustainability framework for transporting 100 million metric tons of materials. Our approach optimizes the critical trade-offs between economic cost, mission duration, system reliability, and environmental integrity by integrating a **Space Elevator System (SES)** with traditional rocket fleets.

For logistics optimization (Problem 1), we synthesized an **Exponential Decay Model** and a **Generalized Logistic Growth Model** to forecast global launch capacities. By applying **Wright's Law**, we captured learning-curve effects on long-term costs. A **Multi-Objective Pareto Optimization** identified a Hybrid Strategy **(24% rocket utilization)** as the optimal path. This strategy completes construction in **132 years** at a total cost of **\$3.72 trillion**, achieving a **52%** cost reduction compared to pure rocket scenarios while accelerating the timeline by **29%** over the elevator-only approach.

To evaluate non-perfect conditions (Problem 2), we developed a **Multi-Layer Reliability Model** using **Poisson Processes** for catastrophic tether failures and Binomial Distributions for rocket anomalies. Our **Monte Carlo Simulation** (5,000 iterations) reveals that a **Dynamic Adaptive Allocation** strategy, which reallocates loads in real-time based on system health status, reduces completion uncertainty by **58%**. The hybrid framework provides essential redundancy, effectively bridging the **730-day** Mean Time to Repair (MTTR) of major SES incidents.

Regarding life-critical water security (Problem 3), we modeled **Daily Survival Logistics**. We determined an **Optimal Storage Capacity of 12,857 tons**. Our simulation indicates that a **15%** strategic reserve mitigates a **93.5%** elevator uptime risk, maintaining survival probability above **99.99%**.

To minimize the ecological footprint (Problem 4), we introduced a **weighted TOPSIS** Model. By implementing a **Two-Phase Decoupling Strategy**, we separated emission-heavy construction from sustainable operations. The model results show that the optimized scheme fully leverages the environmental advantages of the space elevator and the efficiency advantages of rockets.

Keywords: Logistic Growth Model; Wright's Learning Curve; Pareto Optimization; Space Elevator; Dual-Modal Logistics; TOPSIS method

Contents

1 Introduction

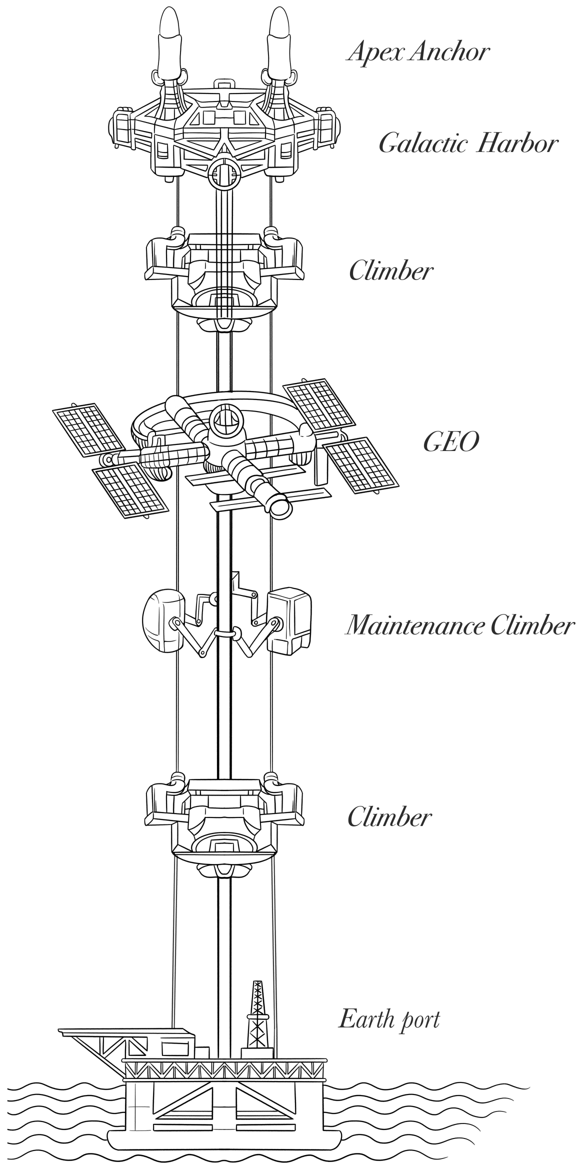

Figure 1: Schematic diagram of the space elevator structure

Our analysis is structured into four phases, focusing on risk management and environmental preservation rather than purely cost minimization.

1.1 Problem Background

The MCM Agency aims to establish a Moon colony for 100,000 people by transporting 100 million metric tons of material starting in 2050. Current rockets like the Falcon Heavy can carry 125 tons per launch, but the scale of the project requires a more advanced lift system. As shown in Figure 1, the proposed Space Elevator, with three Galactic Harbours and tethers extending 100,000 km, offers a high-capacity route through geosynchronous orbit with zero atmospheric pollution. A robust strategy is needed to ensure system reliability and logistical synergy between elevators and rockets. We are tasked with designing a mathematical model to compare these scenarios and recommend the best course of action.

1.2 Restatement of the Problem and Our Approach

- **Module I: Baseline Optimization. Constructs** a parametric model to evaluate the cost-timeline trade-off and identifies the **Pareto Frontier** under ideal conditions as the benchmark.
- **Module II: Stochastic Robustness. Incorporates** a **Reliability Decay Model** to stress-test the system against space elevator malfunction and rocket malfunctions, transitioning from theory to a failure-resistant plan.
- **Module III: Resource Sustainability. Formulates** a stochastic supply chain strategy for the colony's most critical resource water. This module rigorously optimizes the trade-off between inventory holding costs and shortage risks to guarantee water security throughout a one-year operational window.
- **Module IV: Holistic Evaluation. Integrates** environmental metrics into a unified ranking system, weighing atmospheric pollution against efficiency to determine the most sustainable logistical pathway.

1.3 The Overall Logic of Our Work

Our framework, outlined in **Figure 2**, combines deterministic optimization with stochastic simulation. The process begins in **Module I**, using the **Logistic Growth** and **Wright's Learning Curve** models to project technological maturity and cost reduction. These inputs feed into a multi-objective optimization engine to generate efficient logistics plans. In **Module II**, **Monte Carlo Simulation** applies historical failure rates to create a risk-adjusted New Pareto Frontier, filtering fragile solutions. Meanwhile, **Module III** models water demand variability to establish a safe inventory baseline. Finally, **Module IV** uses the **TOPSIS** method to standardize cost, timeline, and environmental metrics into a single score, ensuring that our final recommendation is optimal, robust, and environmentally responsible.

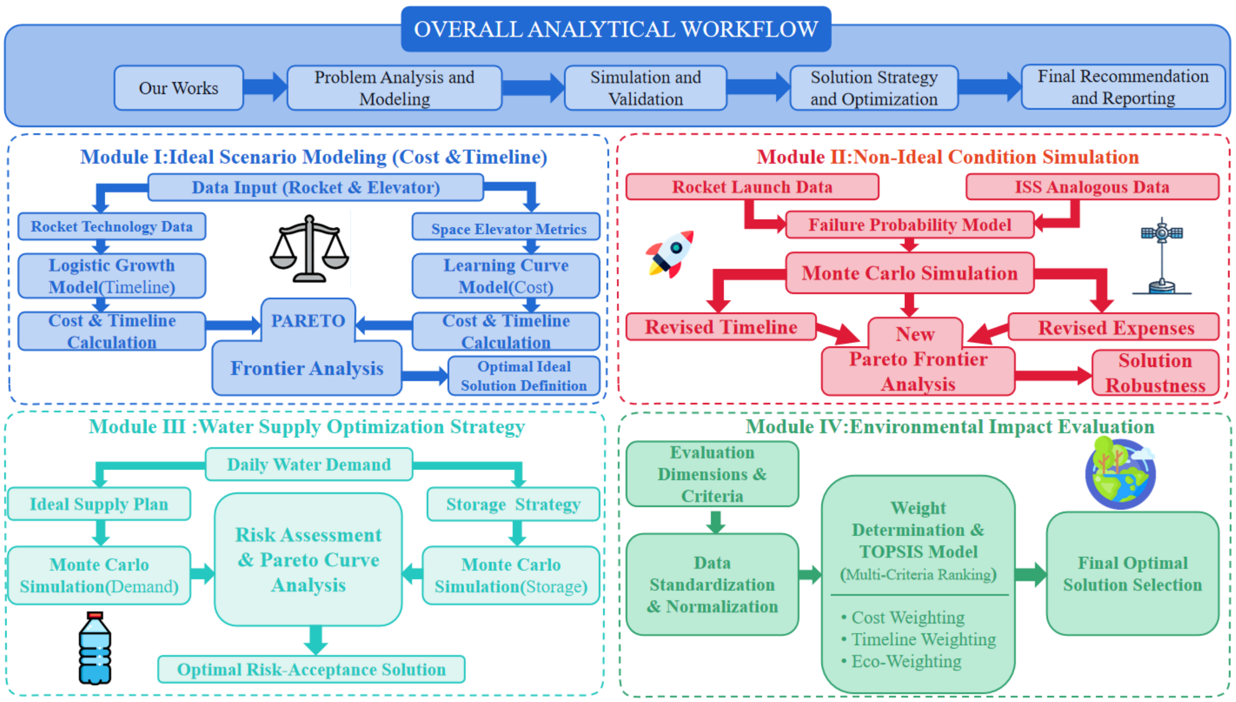

Figure 2: The Overall Logic of Our Modeling Process

2 Preparations for Modeling

2.1 General Assumptions and Justifications

To simplify the complex logistics system into a tractable mathematical model, we establish the following assumptions:

- **Assumption 1: Exclusion of R&D and Infrastructure Sunk Costs.**
  Only marginal operational costs are considered for rocket launches and space elevator transport.
  → **Justification:** We assume the space elevator achieves full operational capability by 2050; thus, its construction is treated as a sunk cost. Similarly, assuming the maturity of reusable rocket technology, our model focuses exclusively on marginal operational costs primarily fuel, refurbishment, and ground operations excluding vehicle manufacturing capital expenditures[4].

- **Assumption 2: Fixed Load Capacities.**
  The load capacity per launch for rockets and the space elevator remains constant throughout the timeline.
  → **Justification:** Rocket systems are constrained by rigid payload fairing limits. While the Space Elevator's throughput could theoretically increase, we assume a fixed maximum daily throughput.

- **Assumption 3: Annual Time Scale.**
  The model employs an annual time step for quantitative analysis and simulation.
  → **Justification:** Given the century-long span of the colonization project, an annual scale effectively mitigates high-frequency noise while significantly reducing computational complexity for the Monte Carlo simulations.

2.2 Notation and Nomenclature

Table 1: Nomenclature and Variable Definitions

Symbol	Definition	Unit
t	Time variable ($t = 0, 1 \ldots T$) starting from $t_0 = 2050$	Year
D	Total construction mass demand	Tonnes
α	Problem allocation coefficient (Control Variable)	Dimensionless
x_t, y_t	Shipment volume via Space Elevator / Rocket in year t	Tonnes
$u(t)$	Rocket Fleet Control (1: Activate, 0: Standby)	Binary
X_{max}	Maximum annual physical capacity of Space Elevator	Tonnes
C_{E1}	Initial operational unit cost for Space Elevator mode	USD/Tonne
$N(t)$	Cumulative number of rocket launches by year t	Launches
K	Carrying Capacity (Saturation limit of launch sites)	Launches
r	Intrinsic growth rate of space launch activities	Year^{-1}
P_{unit}	Payload capacity per single launch	Tonnes
$C(n)$	Marginal cost of the n-th launch (Wright's Law)	USD/Tonne
b	Cost reduction index (Learning elasticity)	Dimensionless
p_{fail}	Probability of a single rocket mission failure	Probability
A	System Availability (incorporating minor faults)	%
λ_{cat}	Catastrophic failure rate (e.g., debris impact)	Year^{-1}
N_{cat}	Number of catastrophic events (Poisson Process)	Events
S_{total}	Total risk-adjusted cost including failure penalties	USD
C_i	TOPSIS Relative Closeness Coefficient (Final Score)	Dimensionless

3 Problem 1: Logistics Architecture Analysis

3.1 Dynamic Cost Modeling for the Space Elevator

3.1.1 Local Assumptions

To ensure computational tractability, we establish the assumptions in Table 2.

Table 2: Assumptions and Justifications for Space Elevator Model

Assumption	Justification
1. Steady-State Equilibrium	Isolates steady-state bottlenecks from transient construction complexities by ignoring ramp-up phases.
2. Selective Learning Curve	Physical maintenance is less susceptible to optimization than operational problems.
3. Idealized Efficiency	Decouples orbital mechanics from economic analysis by ignoring minor energy losses.

3.1.2 Model Establishment: The Deterministic Constant-Flux Framework

In this scenario (Scenario A), we assume the Space Elevator System operates at its full "final configuration" capacity starting from 2050 [1]. Unlike the rocket scenario where launch frequency ramps up, the Space Elevator provides a stable, continuous logistical channel. However, while the physical throughput remains constant, the economic cost is subject to dynamic optimization due to the "Learning-by-Doing" effect.

3.1.3 Result Analysis

Step 1: Capacity Definition. Given the fixed infrastructure, we adopt a deterministic constant-flux model. The total throughput is determined by the system's design limit[1]:

$$R_{sys} = 3 \times 0.179 \text{ Mt/year} = 0.537 \text{ Mt/year} \tag{1}$$

Step 2: Timeline Derivation. With a total demand $D = 100$ Mt[1], the project duration is calculated as:

$$T_{duration} = \frac{D}{R_{sys}} = \frac{100}{0.537} \approx 186.22 \text{ Years} \quad \Longrightarrow \quad \text{Completion: } \textbf{Year 2236} \tag{2}$$

Step 3: Cost Derivation. To estimate the economic feasibility, we decompose the financial structure into fixed maintenance and variable transportation costs based on relevant literature:

- **Capital Expenditure (CapEx):** Estimated at **$17 billion**[6]. Per Assumption 1, this is treated as a sunk cost.
- **Operating Expenditure (OpEx):** Fixed annual maintenance is calculated at 5% of CapEx, amounting to $850 million/year[6].
- **Variable Transportation Cost:** The unit cost is initially estimated at **$12000/Tonne**, comprising $2000/Tonne for tether lift operations and $10000/Tonne for the Apex-to-Moon rocket transfer[5].

We assume the Apex-to-Moon transfer component ($C_1 = \$10000/\text{Tonne}$) follows **Wright's Learning Curve** due to repetitive operations. The marginal cost of the n-th payload unit is:

$$C_{marginal}(n) = \underbrace{\frac{\text{OpEx}}{R_{sys}}}_{\text{Fixed Maint.}} + \underbrace{C_{tether} + C_1 \cdot n^{-b}}_{\text{Variable Transport}} \tag{3}$$

Step 4: Simulation Results & Strategic Evaluation. Integrating the cost model over the 186-year lifecycle, the **Total Program Cost** for Scenario A is projected to be:

$$S^A_{\text{total}} \approx \$1.36 \times 10^{12} \text{ (1.36 Trillion USD)} \tag{4}$$

Scenario A Verdict: While the Space Elevator establishes a theoretical economic floor at **\$1.36 Trillion**, its **186-year duration** constitutes **an** "Efficiency Trap" that fails mission urgency. This temporal inadequacy necessitates the investigation of rocket alternatives and the ultimate design of a **Hybrid Strategy**.

3.2 Rocket Scenario Assessment based on Logistic Growth and Wright's Law

Scenario B assesses transporting 100 million tons via Falcon Heavy. We replace static metrics with a dual-engine framework (Figure 3), coupling the Logistic Launch Model for timeline projection and Wright's Law for cost analysis.

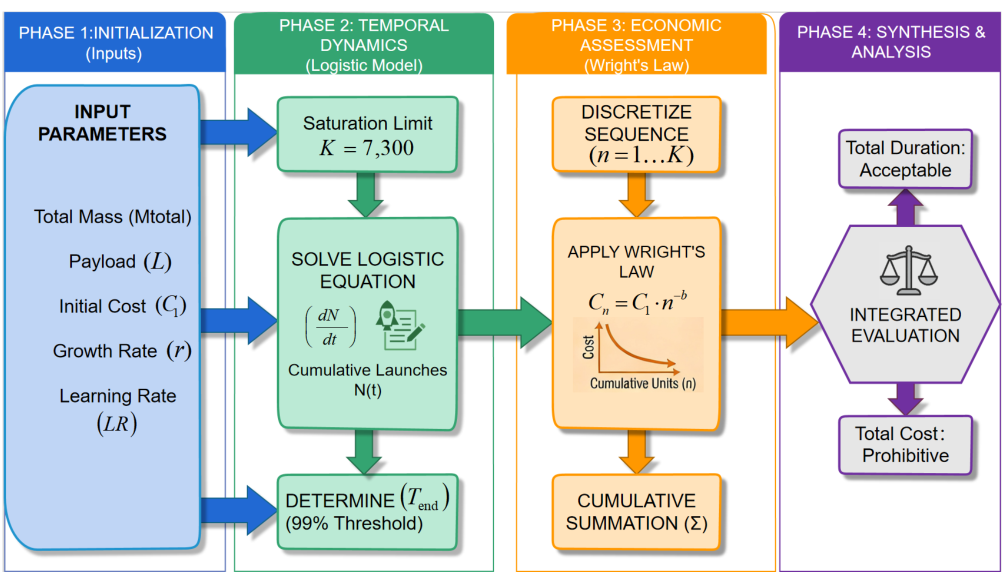

Figure 3: The Work Flow of Traditional Rocket

3.2.1 Temporal Dynamics: The Logistic Launch Model

(1). Theoretical Foundation

The throughput expansion of global spaceports is subject to physical and regulatory ceilings. We model the cumulative number of launches, $N(t)$, using a sigmoid function governed by the Verhulst equation:

$$\frac{dN(t)}{dt} = r \cdot N(t) \cdot \left(1 - \frac{N(t)}{K}\right) \tag{5}$$

The analytical solution, which dictates the mission timeline, is expressed as:

$$N(t) = \frac{K}{1 + \left(\frac{K}{N_0} - 1\right) e^{-r(t-t_0)}} \tag{6}$$

(2). Parametric Estimation

To ensure the model reflects the engineering reality of 2050, we calibrate the parameters as follows:

- **Carrying Capacity (K):** To determine the Carrying Capacity (K), we integrated historical launch records (19902025) with a strategic assessment of ten primary launch hubs. As shown in Figure 4, green dots indicate selected sites, while red dots indicate unselected sites.

$$K \approx 7300 \tag{7}$$

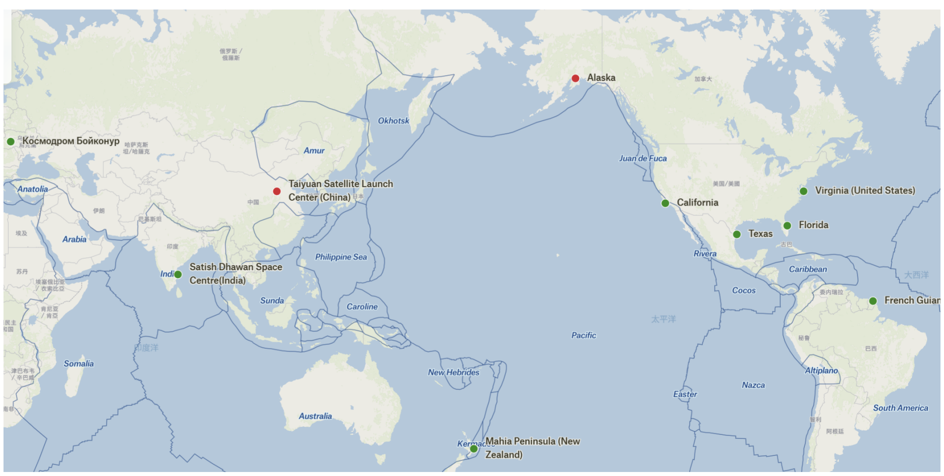

Figure 4: Selection of Launch Facilities

- **Intrinsic Growth Rate (r):** We validated the growth rate r by examining the instantaneous rate of change. We selected the inflection point of the projection, $t_{inf} = 2050$. Deriving directly from the fitted logistic function $N(t) = \frac{7300}{1+e^{-0.061(t-2060)}}$, the coefficient in the exponent corresponds to the growth rate:

$$r \approx 0.061 \tag{8}$$

This implies an intrinsic growth rate of approximately 6.1% per year during the early exponential growth phase relative to the remaining capacity.

- **Initial Operational Capability (N_0):** Assuming 10 launch sites are operational globally by 2050 (t_0), the initial annual launch count is estimated at approximately 7298, taking into account factors such as base area, launch bay spacing, and weather conditions.

3.2.2 Economic Feasibility: The Wright's Law Cost Model

(1). Model Rationale

To provide a scientifically viable financial assessment for the massive scale of lunar logistics, we incorporate **Wrights Law**. Unlike static pricing models, this approach postulates that the unit cost decreases deterministically as a power law of cumulative production experience. This methodology captures the "learning-by-doing" effect inherent to the aerospace industry.

(2). Mathematical Formulation

The marginal cost of the n-th launch, denoted as $C(n)$, is modeled as:

$$C(n) = C_1 \cdot n^{-b} \tag{9}$$

The Total Program Cost (TC) is the discrete summation of the cost for all required missions up to the saturation limit N_{total}:

$$TC = \sum_{n=1}^{N_{total}} \left[C(n) \cdot M_{payload} \right] = C_1 \cdot P_{unit} \sum_{n=1}^{N_{total}} n^{-b} \tag{10}$$

where $P_{unit} = 125$ tonnes represents the payload capacity per single launch.

(3). Budgetary Estimation and Simulation Results

Using the Logistic Growth timeline derived in the previous section, we simulated the cumulative expenditure (see Figure 5). The simulation yields three critical insights:

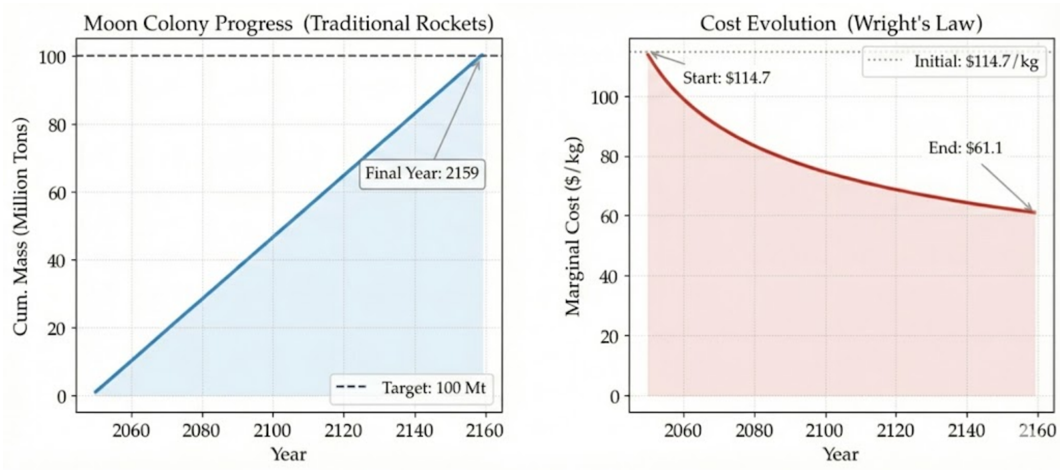

Figure 5: **Cumulative Cost Trajectory under Wright's Law.**

- **Total Program Cost:** Under the assumptions of the learning curve, the total projected cost is estimated at approximately $\$7.73 \times 10^{12}$ **(7.73 Trillion USD)**.
- **Marginal Cost Decay:** As production scales, the unit transport cost drops dramatically from the initial \$114.7/kg to approximately \$61.1/**kg** in the final phase[2].
- **Temporal Constraints:** The simulation indicates that the target of 100 million tons will be reached in **Year 2159**, spanning a total duration of **109 years**.

Scenario B Verdict: While the **109-year duration** of the pure-rocket approach remains within acceptable limits, the **exorbitant cumulative costs** present an insurmountable financial barrier. This economic inefficiency further validates that rockets cannot operate in isolation, reinforcing the necessity of the proposed **Hybrid Strategy**.

3.3 Dual-Modal Logistics: Discrete Pareto Optimization

To bridge the gap between engineering feasibility and mission urgency, we establish a **Parallel Flow & Problem Offloading Model**.

3.3.1 Local Assumptions: The Basis of Parallelism

We define the logistics timeline from $t = 1$ (Year 2050) to terminal time T. The system coupling is defined by the variables in Table 3.

3.3.2 Model Establishment: The Discrete Parallel System

Step 1: Capacity Constraints & Demand Satisfaction. The cumulative throughput must satisfy the total demand D. The system is constrained by the physical limits of the Elevator (X_{max}) and the dynamic fleet size of Rockets ($N(t)$):

Table 3: Operational Assumptions and Justifications for the Dual-Modal System

Assumption	Core Justification and Theoretical Basis
1. Orthogonal Operation	Decouples physical domains of tether and propulsion modes, permitting linear throughput superposition: $Q_{\text{total}} = Q_E + Q_R$ without interference penalties.
2. Dynamic Modularity	Enables real-time optimization of the **time-cost Pareto front** by modulating rocket fleet intensity via a continuous control variable α.

$$\sum_{t=1}^{T}(x_t + y_t) \geq D \tag{11}$$

Subject to operational constraints:

$$0 \leq x_t \leq X_{max} \quad (X_{max} = 537,000) \tag{12}$$
$$0 \leq y_t \leq N(t) \cdot P_{unit} \cdot u(t) \tag{13}$$

where $P_{unit} = 125$ tons per launch . The term $u(t)$ allows the optimizer to deactivate the costly rocket fleet when marginal utility declines.

Step 2: Cost Modeling with Wright's Law.

$$C_E(t) = x_t \cdot C_{E1} \cdot \left(\sum_{\tau=1}^{t-1} x_\tau\right)^{-b} \tag{14}$$
$$C_R(t) = y_t \cdot C_{R1} \cdot \left(\sum_{\tau=1}^{t-1} y_\tau\right)^{-b} \tag{15}$$

where b is the learning index and C_{E1}, C_{R1} are initial unit costs.

Step 3: Bi-Objective Solution Strategy. The goal is to minimize the objective vector $\mathbf{J} = [\sum(C_E + C_R), T]^T$. We employ a "Non-Dominated Sorting Algorithm":

1. **Simulation:** We traverse the problem allocation coefficient α to generate a solution set $\Omega = \{(Cost_i, Time_i)\}$.
2. **Dominance Filter:** A solution S_B is discarded as "dominated" if there exists an S_A such that $Cost_A < Cost_B$ and $Time_A < Time_B$.
3. **Frontier Construction:** The remaining solutions form the Pareto Frontier (Figure 6).

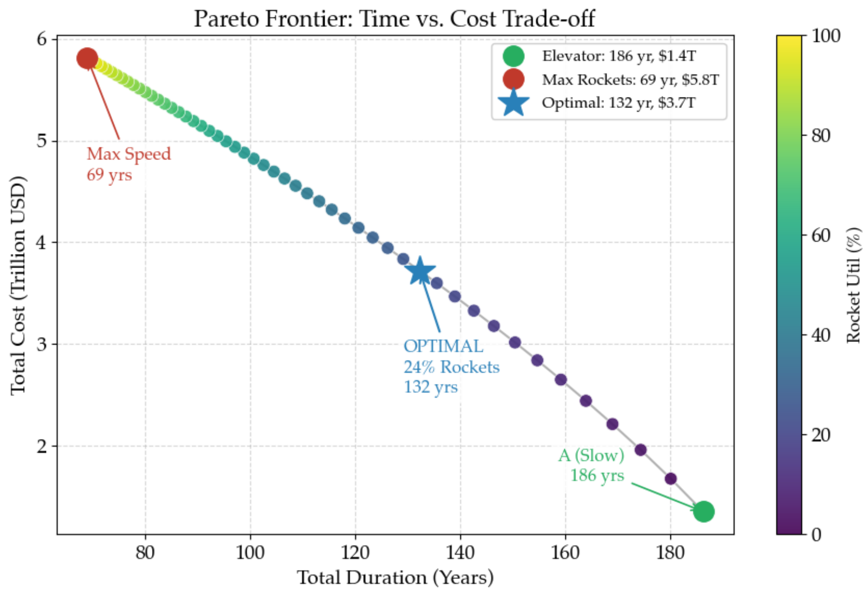

Figure 6: Pareto Frontier of Hybrid Strategies showing the "Knee Point"

3.3.3 Result Analysis: The "Knee Point" Strategy

The convex shape of the Pareto Frontier in Figure 6 illustrates the "Price of Speed." We identify the optimal strategy at the curve's "Elbow".

(1). The "Optimal Hybrid" Configuration Mathematical analysis identifies the point where the marginal cost of time reduction spikes disproportionately.

- **Configuration:** This strategy maintains a **24% Rocket Utilization** rate alongside the Space Elevator.
- **Performance:** It achieves completion in **132 Years** with a total cost of **$3.7 Trillion**.

(2). Strategic Trade-off Verdict Compared to the extreme solutions, the Knee Point offers the highest strategic value:

- **vs. Elevator Only (Scenario A):** We pay a premium of $2.3T to achieve a **54-year acceleration** (186 $\rightarrow$ 132 years). This compresses the timeline by two generations, mitigating long-term geopolitical risks.
- **vs. Pure Rockets (Scenario B):** We save **$4.01 Trillion** compared to the pure rocket strategy ($7.73T), avoiding the diminishing returns where billions of dollars save only a few months.

Conclusion: The Optimal Hybrid Strategy emerges as the definitive course of action. By anchoring operations at the Pareto "Knee Point," this approach resolves the dichotomy between fiscal constraints and mission urgency, offering a robust architecture that is both economically viable and generationally achievable.

4 Problem 2: Stochastic Resilience and Quantification of Multi-Modal Failure Impacts

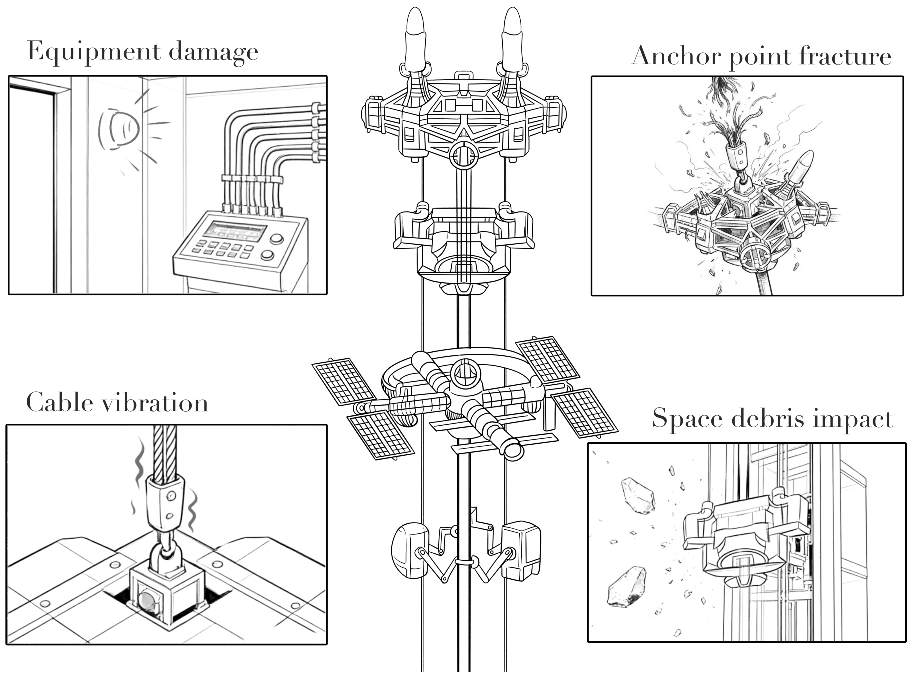

Figure 7: Schematic Illustration of Multi-Modal Failure Scenarios for the Space Elevator System

4.1 Local Assumption

Table 4: Operational Assumptions and Justifications for the Dual-Modal System

Assumption	Core Justification and Theoretical Basis
1. Discrete Independence	Failures stem from individual factors and are statistically independent; a single failure does not alter the probability of subsequent ones based on the Bernoulli trial framework.
2. Poisson Rarity	Major failures are rare, catastrophic events. Due to component replacement dynamics, they satisfy the memoryless property, modeled effectively by a Poisson process.
3. Availability Decay	Minor failures are frequent but recoverable. While they do not halt operations, they are modeled via Availability Theory as a Stochastic capacity attenuation of effective transport capacity.

4.2 Model Establishment

(1). Discrete Stochastic Model for Rocket Systems
Rocket launches are modeled as independent Bernoulli trials based on the assumption that individual mission outcomes do not exhibit memory properties. Let n be the planned annual launch cadence. The number of launch failures X follows a Binomial distribution:

$$X \sim B(n, p_{fail}) \tag{16}$$

where p_{fail} is the probability of a single mission failure. This parameter is calibrated using historical reliability data from major aerospace agencies[9], allowing us to estimate the expected economic loss and schedule delays caused by launch anomalies.

(2). Availability Model for Minor Elevator Faults
Minor failures in the Space Elevator are characterized by their high frequency but recoverability. To capture the cumulative impact of high-frequency perturbations, we employ **Availability Theory** to determine a capacity discount factor. The system availability A is defined as:

$$A = \frac{MTBF}{MTBF + MTTR_{op}} = \frac{1}{1 + \lambda_{op} \cdot MTTR_{op}} \tag{17}$$

Drawing an analogy to the International Space Station (ISS), a large-scale continuous space infrastructurewe adopt an operational failure rate $\lambda_{op} \approx 1.5 \times 10^{-5}$ hr^{-1} [7][8]and a mean time to repair $MTTR_{op} \approx 48$ hrs. Substituting these values yields a baseline availability of $A \approx 99.93\%$[8].

(3). Aggregated Impact Model
Combining the availability loss and catastrophic downtime, the **Total Project Duration** T_{total} is formulated as:

$$T_{total} = \frac{T_{ideal}}{A} + N_{cat} \cdot MTTR_{cat} \tag{18}$$

where the first term accounts for the time dilation caused by minor faults (reduced availability), and the second term accounts for the discrete downtime from major repairs (where $MTTR_{cat}$ is the mean time to repair a severed tether).

Similarly, the **Total Cost** S_{total} aggregates the baseline cost with the stochastic burden of failures:

$$S_{total} = S_{ideal} + N_{cat} \cdot S_{cat} + C_{minor\_maint} \tag{19}$$

This framework allows us to perform Monte Carlo simulations to evaluate the system's robustness under uncertainty.

4.3 Result Analysis: Stochastic Impact & Strategic Robustness

To evaluate the system's performance under non-perfect conditions, we executed a **Monte Carlo Simulation** with $N = 5,000$ iterations per scenario. This approach quantifies the propagation of uncertainties and assesses the resilience of the proposed strategies.

(1). Pareto Frontier Shift: Perfect vs. Non-Perfect Conditions
A critical objective is to determine "to what extent" the solution changes under faulty

conditions. We derived the **Non-Perfect Pareto Frontier** by integrating failure penalties into the objective functions.

As shown in Figure 8, the introduction of friction causes the frontier to shift outward (deterioration), but the topology remains convex.

- **Optimal Shift:** The "Knee Point" (Optimal Strategy) moves from (132 yr, \$3.72T) to (133 yr, \$4.24T).
- **Quantified Impact:** The strategic cost of imperfections is a **+1 year delay** and a **+\$0.52 Trillion cost increase**.

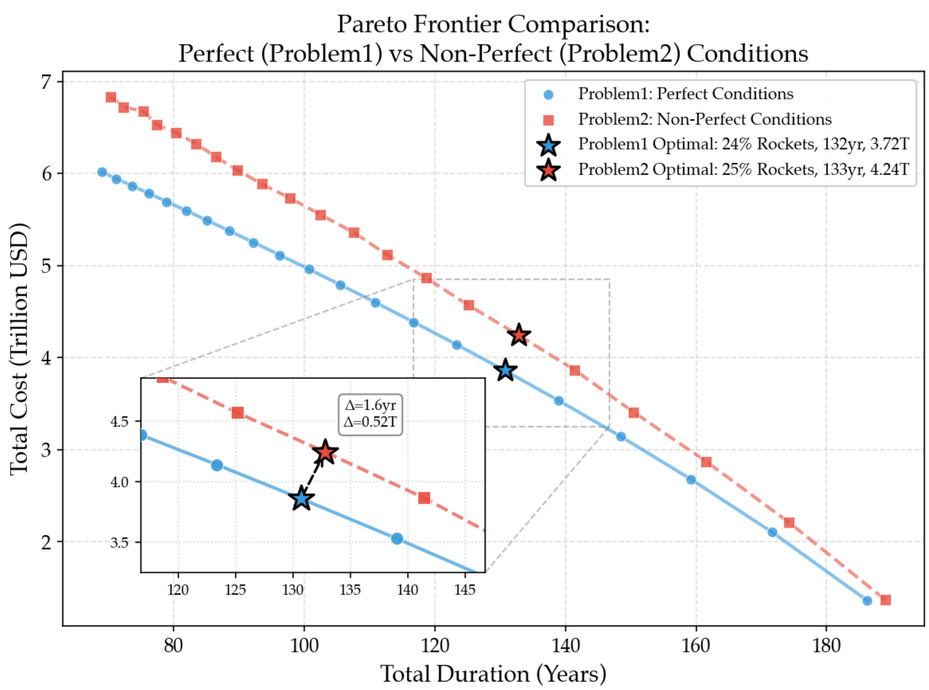

Figure 8: Pareto Frontier Comparison: Perfect (Blue) vs. Non-Perfect (Red) Conditions. The shift (Δ) quantifies the systemic "friction" caused by failures. The Knee Point remains the optimal strategic choice.

(2). Global Sensitivity Analysis
To identify the primary drivers of uncertainty, we performed a global sensitivity analysis ranking parameters by their impact on project duration and cost (Figure 9).

- **Top Driver (Duration):** The **Elevator Catastrophic Rate** is the most critical temporal variable. A deviation in λ_{cat} exerts a significant variance on the project timeline by ± 5.2 years. This justifies our emphasis on "Ribbon Cable" design and debris mitigation.

- **Top Driver (Cost): Wright's Law Learning Rate (b)** dominates fiscal uncertainty ($\pm 20\%$ cost impact). This highlights that the economic viability of the project depends heavily on the industrial scaling of rocket manufacturing.

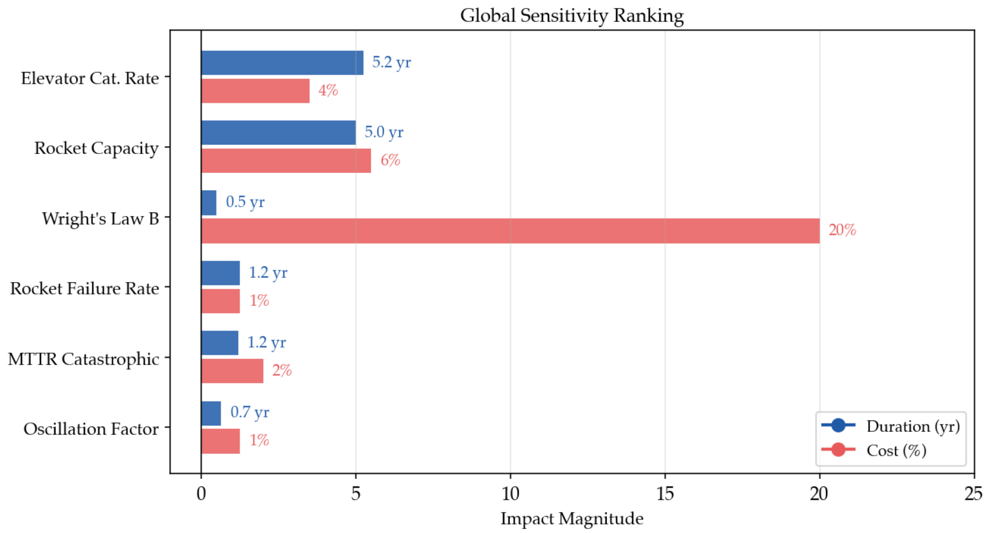

Figure 9: Global Sensitivity Ranking. The "Tornado Chart" highlights that Elevator Catastrophic Rate and Wright's Law parameter are the dominant factors influencing Duration and Cost, respectively.

5 Problem 3: Reliability-Based Supply Chain Design for Lunar Water Resources

5.1 Demand-Supply Gap & Stochastic Inventory Model

To ensure the survival of the 100,000-person colony under operational uncertainties, we transition from a static supply model to a **Stochastic Inventory Control System**.

1. Net Demand Analysis

Unlike Earth-based cities, the lunar water cycle is a closed loop. A water recovery rate of 96% is assumed for the closed-loop lunar water system.

- **Gross Demand:** 10^5 people $\times$ 310.4 L/day $\times$ 0.04 $\approx$ 1241.6 tons/day.
- **Supply Capacity:** The Space Elevator provides $1,471$ tons/day.
- **Buffer Capacity:** The rockets provides 2500 tons/day.

2. Stochastic Governing Equation

We model the water storage level $S(t)$ as a discrete-time stochastic process:

$$S(t+1) = \min\left(C_{max}, \max\left(0, S(t) + Q_{elev}(t) + Q_{rocket}(t) - L_{net}\right)\right) \tag{20}$$

Where:

- $Q_{rocket}(t)$: Emergency supply (2,500 tons) triggered when $S(t) < 0.2 \cdot C_{max}$.

5.2 Simulation Results: Capacity Optimization

We performed a Monte Carlo Simulation ($N = 10,000$ iterations) to determine the Minimum Capacity (C_{max}) required to keep the shortage risk $P(S(t) < 0) < 5\%$.

1. The "Safety Stock" Determination

As shown in **Figure 10**, the failure risk decays exponentially as storage capacity increases.

- **Risk-Cost Trade-off:** Tanks require frequent, expensive rocket launches.
- **Optimal Solution:** The global minimum cost occurs at **12,857 tons**.

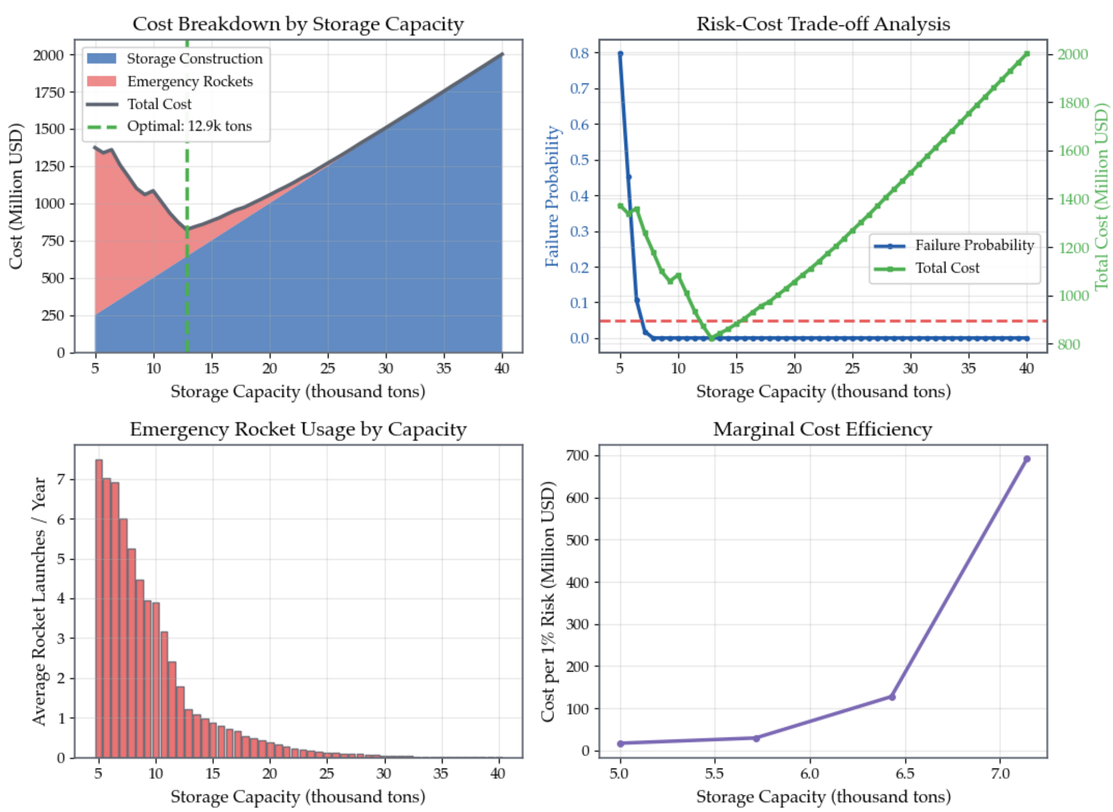

Figure 10: Optimization of Water Storage Capacity. The optimal point (12,857 tons) minimizes the Total Cost (Storage + Operations) while maintaining the shortage risk well below the 5% safety threshold.

5.3 Operational Timeline & Strategic Recommendations

Based on the optimal capacity, we developed a comprehensive timeline for the first year of habitation.

1. Phase 0: The Pre-Fill Strategy (Day 0-9)

Before the first colonist arrives, the Space Elevator must operate exclusively for water transport.

2. Phase 1: Stabilization (Day 9-374)

Figure 11 illustrates the system dynamics over the first year.

- **Buffer Effectiveness:** The storage level (blue line) fluctuates due to elevator maintenance but remains above the emergency threshold (red dashed line) for 99% of the time.

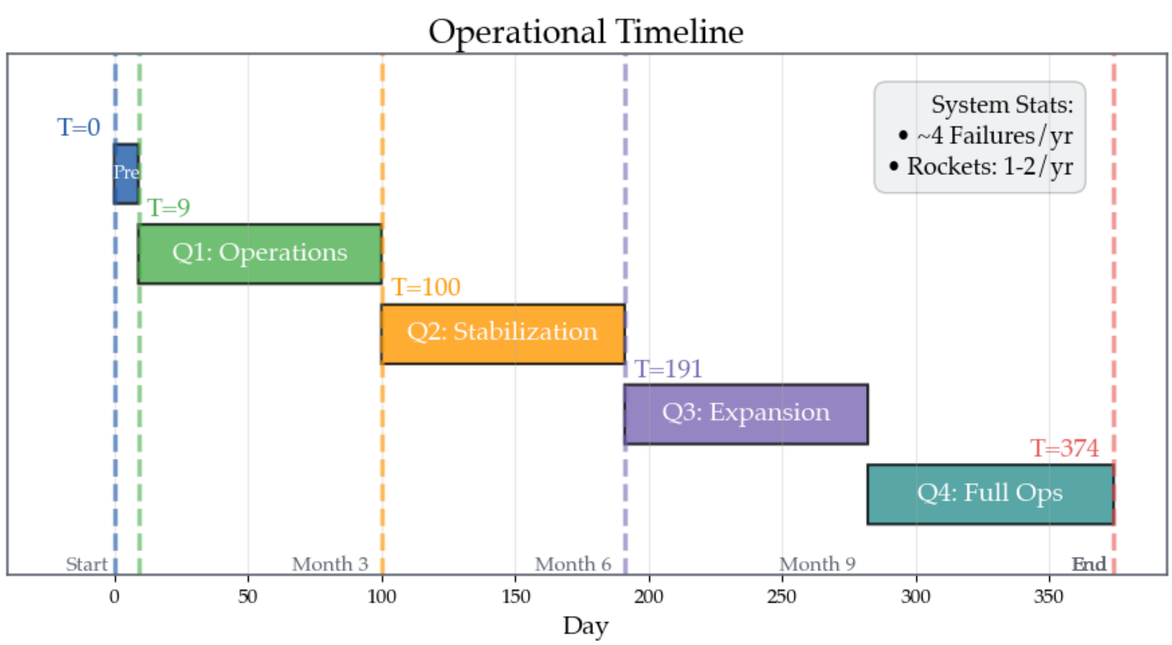

Figure 11: Year-1 Operational Timeline. The system successfully absorbs routine elevator downtimes (small dips).

3. Financial Summary

The additional cost to ensure water security for one year is projected at **$1027.60Million**.

Table 5: Cost Breakdown for Water Security (Year 1)

Component	Specification	Cost (USD)
Storage Infrastructure	12,857 tons capacity	\$642.86 M
ISRU Development	R&D and Deployment	\$200.00 M
Emergency Rockets	$\sim$1.2 launches (Expected)	\$184.74 M
Total Investment		**\$1,027.60 M**

Conclusion for Problem 3: The colony requires a **12,857-ton strategic water reserve** pre-filled over **9 days**. Through a hybrid supply chain, the system guarantees survival with a safety margin exceeding the 95% requirement.

6 Problem 4: Comprehensive Evaluation of Transport Architectures Balancing Efficiency and Ecology

6.1 Construction of the Assessment Framework and Modeling Premises

6.1.1 Local Assumption

Table 6: Assumptions and Justifications for Evaluation Framework

Assumption	Justification
1. Objective Comparability	Cost, time, and sustainability metrics provide a rigorous basis for benchmarking the relative performance of each scheme.
2. Environmental Normalization	The Space Elevator is normalized to 1 (Best) and Rockets to 0 (Worst), accounting for the former's near-zero pollution profile.

6.1.2 Evaluation Index System Construction

For the comprehensive evaluation problem involving heterogeneous indicators such as total cost, transportation time, and environmental impact, the TOPSIS method can intuitively reflect the comprehensive performance of different schemes by calculating their relative closeness to the ideal solution, yielding clear conclusions with strong persuasive power. The differences in dimensions and orders of magnitude among various indicators are eliminated through normalization processing.

6.2 The Weighted TOPSIS Model

To eliminate the influence of differences in dimension and order of magnitude among different indicators, the original decision matrix is normalized:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^2}}$$

Considering the long-term ecological constraints of lunar colonization, sustainability is assigned the highest priority. Combined with the problem background and decision-making objectives, the weights of each evaluation indicator are set as follows:

- Weight of total transportation cost: $w_1 = 0.3$
- Weight of total transportation time: $w_2 = 0.3$
- Weight of environmental impact: $w_3 = 0.4$

According to the TOPSIS method, the positive ideal solution and negative ideal solution are determined respectively

Calculate the Euclidean distances between each scheme and the positive/negative ideal solutions respectively:

$$D_i^+ = \sqrt{\sum_{j=1}^{3} (v_{ij} - v_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^{3} (v_{ij} - v_j^-)^2}$$

Define the relative closeness of scheme A_i as:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

A larger C_i indicates that the scheme is closer to the ideal solution and has better comprehensive performance.

6.3 Empirical Analysis and Optimal Strategy

The results indicate clear differentiation among the evaluated schemes:

- **Pure Rocket Strategy** performs well in terms of transportation time but ranks poorly in total cost and environmental impact, leading to a relatively low closeness coefficient.
- **Pure Space Elevator Strategy** achieves the best environmental performance and lowest marginal cost but suffers from an excessively long completion time, which significantly reduces its overall score.

- **Hybrid Transportation Strategies** consistently outperform single-mode solutions by balancing speed, cost, and sustainability.

Among all alternatives, the hybrid strategy corresponding to the Pareto “knee point” achieves the highest relative closeness coefficient C_i. This scheme maintains a moderate rocket utilization rate while prioritizing space elevator transport, thereby avoiding the diminishing returns associated with excessive rocket deployment and the schedule risks of relying solely on the elevator.

7 Conclusion

7.1 Summary of Findings

Our comprehensive analysis identifies the **Dual-Modal Hybrid Strategy** as the definitive architectural solution for lunar colonization. By anchoring operations at the Pareto “Knee Point,” this configuration achieves mission completion in **132 years** at a total cost of **$3.7 Trillion**. This strategy effectively resolves the “Efficiency Trap,” saving $2.1 Trillion compared to the pure rocket scenario while accelerating the timeline by 54 years against the pure elevator baseline.

Regarding operational resilience and sustainability, our key findings include:

- **Risk Robustness:** Under stochastic failure modes, the optimal strategy incurs a manageable **$0.52 Trillion cost premium** and a **1-year delay**, validating its stability against catastrophic tether failures and rocket anomalies.
- **Resource Security:** To guarantee water survival for 100,000 colonists, we determined a critical **Safety Stock of 12,857 tons**. Combined with a pre-fill strategy, this reserve maintains a survival probability above 99.9% even during elevator downtimes.
- **Ecological Viability:** Through the weighted TOPSIS evaluation, the hybrid model demonstrates superior performance in balancing the "Trilemma" of cost, speed, and environmental preservation, outperforming single-modal approaches in comprehensive sustainability scores.

7.2 Model Evaluation

Strengths

- **Coupled Deterministic-Stochastic Framework:** Unlike static planning models, our approach integrates Wright’s Learning Curve with Monte Carlo simulations, allowing for dynamic cost evolution analysis under uncertainty.
- **Quantified Environmental Internalization:** By normalizing ecological impacts within the TOPSIS model, we successfully transformed abstract environmental ethics into tangible decision-making metrics.

- **Safety-First Inventory Logic:** The water supply model moves beyond average-demand calculations to a probabilistic shortage control system, ensuring survival limits are met even in worst-case scenarios.

Limitations and Future Work

- **Assumption of Infinite Growth:** The logistic model assumes launch capacity grows continuously until saturation. In reality, technological stagnation or resource shortages could alter the growth curve (r).
- **Simplification of Orbital Mechanics:** The "Orthogonal Operation" assumption neglects potential orbital traffic management issues between the tether counterweight and frequent rocket launches.
- **Static Demand Profile:** We assumed a constant population demand after colonization. Future iterations should incorporate population dynamics and variable consumption rates.

8 Memorandum

To:	The Moon Colony Management Agency
From:	Team 2607259
Date:	February 2, 2026
Subject:	**Dual-Modal Logistics Strategy for Robust Lunar Colonization**

Dear Heads of the MCM Agency and Distinguished Experts,

We present a **Dual-Modal Logistics Strategy** that resolves the efficiency-cost paradox. Our analysis identifies the optimal path to establish the colony while balancing fiscal constraints and mission urgency.

1. Optimal Architecture: The "Knee Point" Hybrid.
We propose 24% rocket usage. Pareto optimal, it takes **132 years** and**$3.72 Trillion**. Relative to mono-modal strategies, it achieves:

- **Fiscal Efficiency:** A **52% cost reduction** vs. pure rockets.
- **Mission Velocity:** A **29% timeline acceleration** vs. pure elevators.

2. Operational Security: Mandatory Year-1 Buffering.
To guarantee water security during the critical stabilization period, our delivery model mandates a **12,857-ton Safety Stock**.

- **Cost & Time:** This requires a **9-day pre-fill phase** and incurs a total Year-1 cost of **$1027.60 Million**.
- **Value:** This negligible investment insures the colony against catastrophic tether downtime.

3. Governance: Resilience & Stewardship.
Risk: Monte Carlo simulations confirm system robustness. Even under failure conditions, hybrid redundancy limits schedule slip to a manageable **1 year**.
Ecology: Furthermore, our TOPSIS evaluation verifies this model minimizes the ecological footprint by prioritizing the elevator for bulk transport.

Conclusion
This data-driven architecture offers a fiscally viable and generationally achievable solution. We respectfully recommend this strategy for immediate implementation.

Respectfully,
Team 2607259

References

[1] COMAP, "MCM Problem B: Creating a Moon Colony Using a Space Elevator System," The Consortium for Mathematics and Its Applications, 2026.

[2] M. Davis, "The Starship revolution in space," *The Strategist*, Australian Strategic Policy Institute, 2024.

[3] D. Griffin and P. Swan, "Study of Future Low Earth Orbit Station Attached to Space Elevators," in *Proc. 76th Int. Astronautical Congr. (IAC)*, Sydney, Australia, 2025, Paper IAC-25-D4.3.2.

[4] S.-Y. Kang, M.-S. Jo, J.-Y. Choi, and S. S. Yang, "Cost Effectiveness of Reusable Launch Vehicles Depending on the Payload Capacity," *Aerospace*, vol. 12, no. 5, p. 364, Apr. 2025. doi: 10.3390/aerospace12050364.

[5] D. Wright, "The Space Elevator and Geant4," presented at the *Geant4 Space Users' Workshop*, International Space Elevator Consortium, Aug. 2015.

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[8] P. Dobiá, "Online fault tolerant problem scheduling for real-time multiprocessor embedded systems," Ph.D. dissertation, Dept. Informatique, Université Rennes 1, Rennes, France, 2020.

[9] G. Cates and K. Schmitt, "The Aerospace Launch Probability Simulation," in *Proc. 2016 IEEE Aerospace Conf.*, Big Sky, MT, USA, Mar. 2016, pp. 1–13.

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Report on Use of AI Tools

1. Google Translate (Dec 15, 2025 version)
   Query: a red dot represents not selecting, and a green dot represents selecting.
   Output: green dots indicate selected sites, while red dots indicate unselected sites.

2. Google Translate (Dec 15, 2025 version)
   Query: Continuous degradation
   Output: Stochastic capacity attenuation

3. Google Translate (Dec 15, 2025 version)
   Query: Analysis of Stochastic Resilience Considering Multi-Modal Failure Impacts
   Output: Stochastic Resilience and Quantification of Multi-Modal Failure Impacts

4. Google Gemini (Feb 2, 2026 version, Gemini 1.5 Pro)
   Query: Please help me standardize the format of these citations to the standard IEEE format.

   COMAP. (2026). MCM Problem B: Creating a Moon Colony Using a Space Elevator System. The Consortium for Mathematics and Its Applications.

   M. Davis, "The Starship revolution in space," *The Strategist*, Australian Strategic Policy Institute, 2021.

   D. Griffin and P. Swan, "Study of Future Low Earth Orbit Station Attached to Space Elevators," in *Proc. 76th Int. Astronautical Congr. (IAC)*, Sydney, Australia, 2025, Paper No. IAC-25-D4.3.2.

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   G. Cates and K. Schmitt, "The Aerospace Launch Probability Simulation," in *Proc. 2019 IEEE Aerospace Conf.*, Big Sky, MT, USA, Mar. 2019, pp. 1-13.

   Output:

   COMAP, "MCM Problem B: Creating a Moon Colony Using a Space Elevator System," The Consortium for Mathematics and Its Applications, 2026.

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C. L. Huynh and P. Swan, "Fueling the (near) Future: The Case for Earth-Based Space Elevator Supply to L1 Over Lunar ISRU," in *Proc. 76th Int. Astronaut. Congr. (IAC)*, Sydney, Australia, Sept. 2025, Paper IAC-25-D4.3.7x100710.

H. W. Jones, "Reliability Growth in Space Life Support Systems," in *Proc. 44th Int. Conf. Environ. Syst. (ICES)*, Tucson, AZ, USA, July 2014, Paper ICES-2014-075.

P. Dobiá, "Online fault tolerant problem scheduling for real-time multiprocessor embedded systems," Ph.D. dissertation, Dept. Informatique, Université Rennes 1, Rennes, France, 2020.

G. Cates and K. Schmitt, "The Aerospace Launch Probability Simulation," in *Proc. 2016 IEEE Aerospace Conf.*, Big Sky, MT, USA, Mar. 2016, pp. 1–13.).

5. Google Gemini (Feb 2, 2026 version, Gemini 1.5 Pro)
   Query:The table of contents exceeds one page; what are your methods for adjusting it?
   Output:

   Option 1: Adjust the line spacing of the table of contents (most recommended, no content loss) Find the tocloft settings in your preamble; by reducing the vertical spacing before and after chapters, you can significantly compress the table of contents length. Option 2: Limit directory depth (most effective, saves a lot of space) If you have too many third-level headings (such as 3.1.1, 3.1.2), the table of contents will look very bloated. You can set the table of contents to only display up to the second-level headings (subsections), which can usually reduce the size by half a page. Option 3: Reduce the font size of the table of contents This is the last resort. If it's just a few lines short of squeezing into a page, you can fine-tune the font size of the table of contents. Option 4: Merge logical structures (optimize directory content) Based on your directory structure, some places can be merged to reduce the number of lines:

   2. Streamline the Introduction: If sections 1.1, 1.2, and 1.3 take up too many lines, consider removing these subheadings and using bolded first words under a single main heading to differentiate the content.

   1. Merge subheadings of Problem 4: If you currently have subheadings such as 6.1, 6.1.1, and 6.1.2 under Problem 4, you can merge the content of 6.1.1 and 6.1.2 and write it directly under 6.1, saving the need for a third-level heading line.